ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** negative charge	Negative.
2	**(b)** 2 electrons	2n squared.
3	**(d)** atomic number	Z.
4	**(a)** positive charge	Positive.
5	**(c)** 8 electrons	2 x 4.
6	**(c)** mass number	A.
7	**(c)** no charge	Neutral.
8	**(b)** 2, 4	C.
9	**(a)** three isotopes	Protium, deuterium, tritium.
10	**(a)** protons and neutrons	Nucleons.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Electron (-), proton (+) (1); Neutron (0) (1).	1+1
12	Combining capacity of an element (2).	2
13	Z = 11 (1); 2, 8, 1 (1).	1+1
14	Thomson (1); Rutherford (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Most alpha particles passed (1); Some were deflected (1); Few rebounded (1).	1+1+1
16	Be 2, 2 (1); B 2, 3 (1); C 2, 4 (1).	1+1+1
17	Medicine (1); Nuclear energy (1); Research (1).	1+1+1
	OR: Atom is mostly empty (1); Nucleus is positive (1); Nucleus is small and dense (1).	1+1+1
18	F 2, 7 (1); Ne 2, 8 (1); Na 2, 8, 1 (1).	1+1+1
	OR: No nucleus (1); Could not explain alpha scattering (1); No electron shells (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Positive sphere (1); Electrons embedded (1); Plum pudding (1); Neutral atom (1); Drawback (1).	1+1+1+1+1
	OR: Definition (1); Hydrogen (1); Chlorine (1); Uses (1); Same chemical properties (1).	1+1+1+1+1
20	Nucleus (1); Electrons revolve (1); Empty space (1); Positive nucleus (1); Drawback (1).	1+1+1+1+1
	OR: Definition (1); Ar and Ca (1); Same mass number (1); Different Z (1); Different chemical properties (1).	1+1+1+1+1

Mistakes to avoid:

- Confusing atomic number with mass number, or finding neutrons as $A + Z$ instead of $A - Z$.
- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.
- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2, 6	O.
2	**(b)** cobalt-60	Gamma rays.
3	**(d)** protons	Z.
4	**(d)** 2, 8	Ne.
5	**(c)** iodine-131	Thyroid.
6	**(b)** protons + neutrons	A.
7	**(b)** 1	2, 8, 1.
8	**(c)** uranium-235	Fission.
9	**(c)** nucleus	Gold foil.
10	**(c)** 2	2, 6.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Same atomic number, different mass numbers (2).	2
12	Cobalt-60: cancer treatment (2).	2
13	Number of protons in the nucleus (2).	2
14	Same mass number, different atomic numbers (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	p = 6 (1); e = 6 (1); n = 6 (1).	1+1+1
16	H = 1 (1); Li = 1 (1); F = 1 (1).	1+1+1
17	Electrons in shells (1); No radiation in shells (1); Energy changes when jumping (1).	1+1+1
	OR: p = 8 (1); e = 8 (1); n = 8 (1).	1+1+1
18	Mg = 2 (1); Ca = 2 (1); O = 2 (1).	1+1+1
	OR: A = 35 (1); 2, 8, 7 (1); Chlorine (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Shells (1); No radiation (1); Energy levels (1); K, L, M, N (1); Jump (1).	1+1+1+1+1
	OR: Definition (1); Na: 1 (1); Mg: 2 (1); O: 2 (1); Ne: 0 (1).	1+1+1+1+1
20	Set-up (1); Observation 1 (1); Observation 2 (1); Observation 3 (1); Conclusion (1).	1+1+1+1+1
	OR: C: 6, 6, 6 (1); N: 7, 7, 7 (1); O: 8, 8, 8 (1); F: 9, 9, 10 (1); p, e, n listed (1).	1+1+1+1+1

Mistakes to avoid:

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- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.
- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 12	Sum.
2	**(c)** neutron	1932.
3	**(b)** 0	Complete shell.
4	**(d)** 8	14 - 6.
5	**(a)** electron	Cathode rays.
6	**(a)** 6	Z = 6.
7	**(a)** neutral	p = e.
8	**(b)** shells (orbits)	Energy levels.
9	**(a)** 8	Z = 8.
10	**(c)** mass	1 u.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Eight electrons in the outer shell (2).	2
12	Sum of protons and neutrons (2).	2
13	N has Z = 7 (1); 2, 5 (1).	1+1
14	Niels Bohr (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	p = 11 (1); e = 11 (1); n = 12 (1).	1+1+1
16	Protium 1 (1); Deuterium 2 (1); Tritium 3 (1).	1+1+1
17	Hydrogen (1); Carbon (1); Oxygen (1).	1+1+1
	OR: H 1 (1); He 2 (1); Li 2, 1 (1).	1+1+1
18	Charge (1); Mass (1); Location (1).	1+1+1
	OR: Sodium (1); Magnesium (1); Chlorine (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	H, He (1); Li, Be (1); B, C (1); N, O (1); F, Ne (1).	1+1+1+1+1
	OR: Electron (1); Proton (1); Neutron (1); Charges (1); Masses (1).	1+1+1+1+1
20	Z = 11 (1); A = 23 (1); n = 12 (1); p = e = 11 (1); Notation (1).	1+1+1+1+1
	OR: Eight electrons (1); He has 2 (1); Complete shells (1); Unreactive (1); Examples (1).	1+1+1+1+1

Mistakes to avoid:

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- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
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